IP Address Analysis And Reporting

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Executive Summary

This lab focused on exploring ip address classification and types, subnetting and address capacity of selected targets.key activities includes analyzing real-world websites, classifying their IP addresses, calculating device capacities and determining subnet masks. The lab revealed the category that the ip address of selected targets falls to mostly class A and B and the capacity of host they can occupied.

1.0 Introduction

IP address analysis in IPv4 focuses on understanding how a 32-bit address is structured and how it is divided into network and host portions. An IPv4 address is written in dotted-decimal form, such as 192.168.1.10, and consists of four octets. Part of the address identifies the network to which a device belongs, while the remaining part identifies the specific host within that network. By examining the subnet mask, one can determine where the network portion ends and where the host portion begins, which is essential for routing, subnetting, and efficient IP address management.

IPv4 analysis also involves recognizing the traditional IP address classes. Class A addresses use the first octet for the network and the remaining three for hosts, supporting very large networks. Class B uses the first two octets for the network and the last two for hosts, suitable for medium-sized networks. Class C uses the first three octets for the network and the last one for hosts, designed for smaller networks. Classes D and E are reserved for multicast and experimental purposes. Understanding these classes and the separation of network and host portions forms the foundation for subnetting, routing decisions, and overall network design.

2.0 Aim and objectives

The aim of this lab is to understand the concept of IP addressing, network scalability, and subnetting.

The Objectives of this lab is to

- analyze real-world websites,
- classify IP addresses,
- calculate device capacities
- determine subnet masks, and
- perform advanced IP operations.

2.1 Tools

- Kali Linux

3.0 Methodology

3.1 Selected Websites of Interest

- Amazon.com - 98.87.170.74
- Netflix.com - 54.246.79.9
- Yahoo.com - 74.6.143.26
- Coursera.com - 52.72.124.35
- Kaspersky.com - 77.74.178.24
- Youtube.com - 142.251.216.46
- Facebook.com - 102.132.101.35
- X.com - 162.159.140.229
- LinkedIn.com - 150.171.22.12
- Wikipedia.org - 185.15.58.224

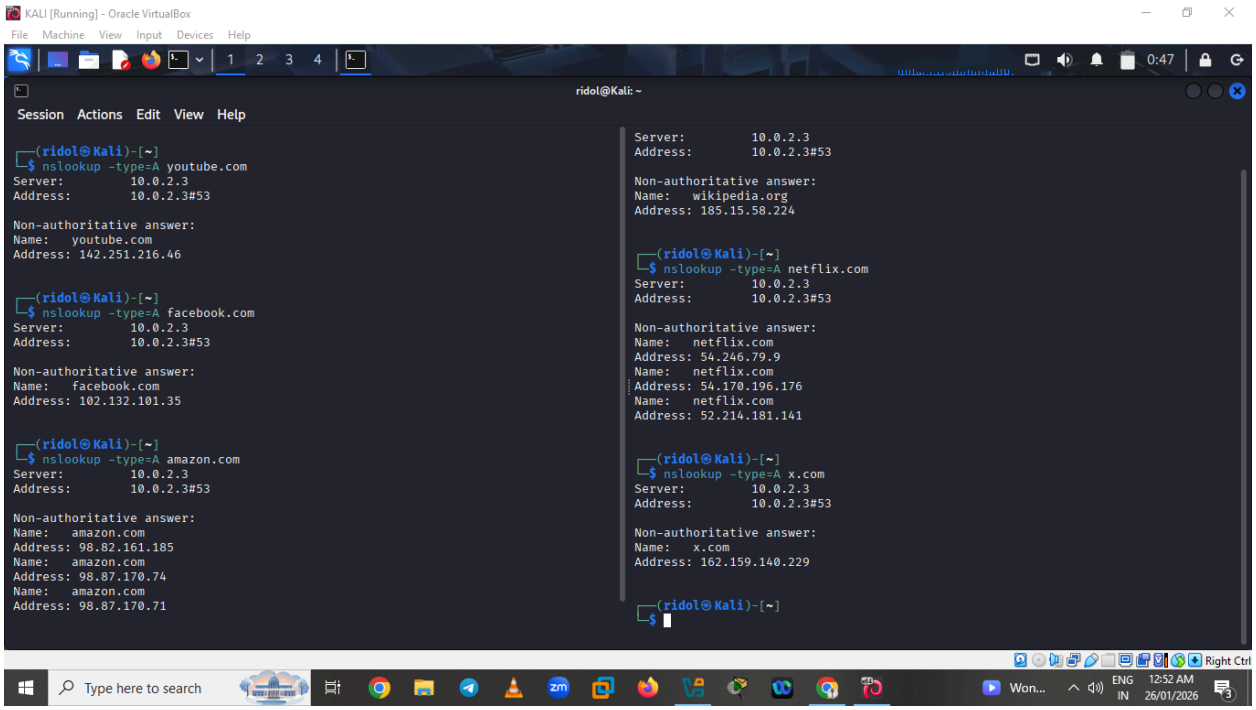

Fig 1: Evidence showing the ip address of selected target from nslookup queries

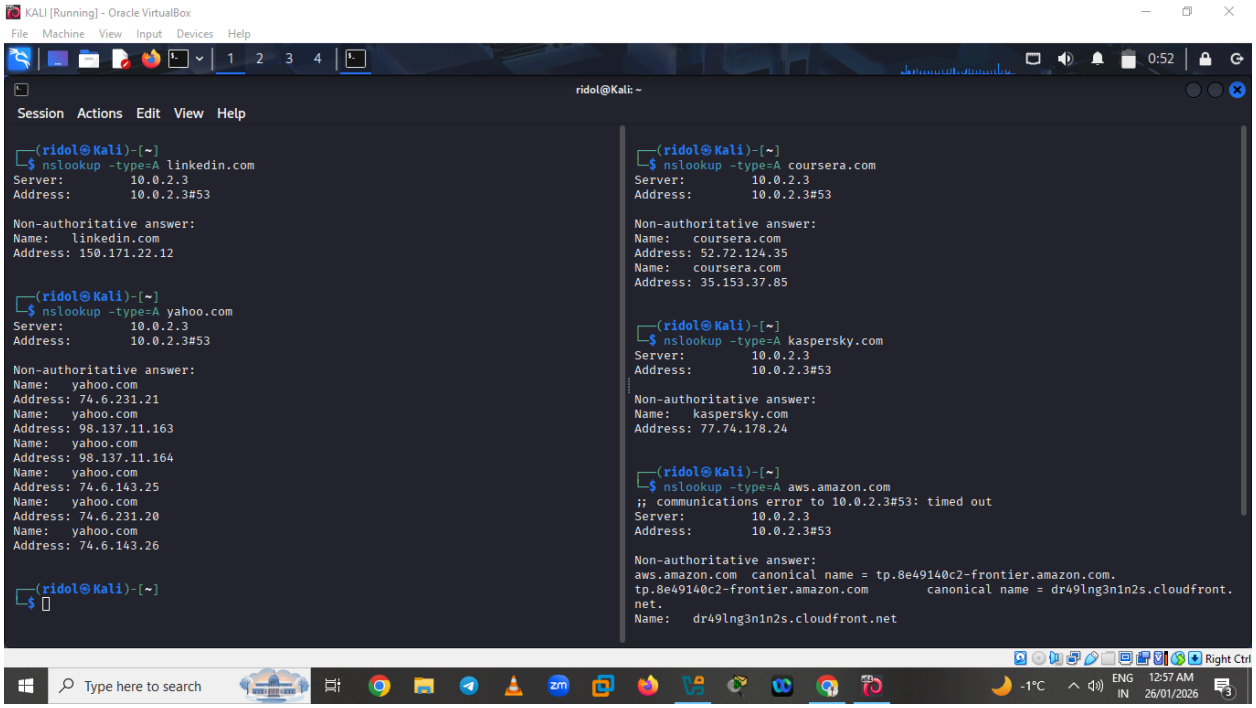

Fig 2: Evidence showing the ip address of selected target from nslookup queries

3.2 IP Address Classification

Domain	ip Address	subnet Mask	class
Amazon.com	98.87.170.74	255.0.0.0	A
Netflix.com	54.246.79.9	255.0.0.0	A
Yahoo.com	74.6.143.26	255.0.0.0	A
Coursera.com	52.72.124.35	255.0.0.0	A
Kaspersky.com	77.74.178.24	255.0.0.0	A
Youtube.com	142.251.216.46	255.255.0.0	B

Facebook.com	**102.132.101.35**	**255.0.0.0**	**A**
X.com	**162.159.140.229**	**255.255.0.0**	**B**
LinkedIn.com	**150.171.22.12**	**255.255.0.0**	**B**
Wikipedia.org	**185.15.58.224**	**255.255.0.0**	**B**

3.2 IP Address Analysis

Amazon.com - **98.87.170.74**

1. IP Class : First octet: 98 (**Class A)**

2. Default Subnet Mask : 255.0.0.0 (/8)

3. Network (Subnet) Address : 98.0.0.0

4. **Broadcast Address** : 98.255.255.255

5. **Host Portion :** 87.170.74

6. **Number of Usable Hosts**

- Class A has 24 host bits:
- 2^24 − 2 =16,777,214 usable addresses
- **First Usable Host :** 98.0.0.1
- **Last Usable Host** : 98.255.255.254

Netflix.com - **54.246.79.9**

1. IP Class : First octet: 54 (**Class A)**

2. Default Subnet Mask : 255.0.0.0 (/8)

3. Network (Subnet) Address : 54.0.0.0

4. **Broadcast Address** : 54.255.255.255

5. **Host Portion :** 246.79.9

6. **Number of Usable Hosts**

- Class A has 24 host bits:
- 2^24 − 2 =16,777,214 usable addresses
- **First Usable Host :** 54.0.0.1
- **Last Usable Host** : 54.255.255.254

Yahoo.com - 74.6.143.26

1. IP Class : First octet: 74 (**Class A)**

2. Default Subnet Mask : 255.0.0.0 (/8)

3. Network (Subnet) Address : 74.0.0.0

4. **Broadcast Address** : 74.255.255.255

5. **Host Portion :** 6.143.26

6. **Number of Usable Hosts**

- Class A has 24 host bits:
- 2^24 − 2 =16,777,214 usable addresses
- **First Usable Host :** 74.0.0.1
- **Last Usable Host** : 74.255.255.254

Coursera.com - 52.72.124.35

1. IP Class : First octet: 52 (**Class A)**

2. Default Subnet Mask : 255.0.0.0 (/8)

3. Network (Subnet) Address : 52.0.0.0

4. **Broadcast Address** : 52.255.255.255

5. **Host Portion :** 72.124.35

6. **Number of Usable Hosts**

- Class A has 24 host bits:
- 2^24 − 2 =16,777,214 usable addresses
- **First Usable Host :** 52.0.0.1
- **Last Usable Host** : 52.255.255.254

Kaspersky.com - 77.74.178.24

1. IP Class : First octet: 77 (**Class A)**

2. Default Subnet Mask : 255.0.0.0 (/8)

3. Network (Subnet) Address : 77.0.0.0

4. **Broadcast Address** : 77.255.255.255

5. **Host Portion :** 74.178.24

6. **Number of Usable Hosts**

- Class A has 24 host bits:
- 2^24 − 2 =16,777,214 usable addresses
- **First Usable Host :** 74.0.0.1
- **Last Usable Host** : 74.255.255.254

Youtube.com - 142.251.216.46

1. IP Class : First octet: 142 (**Class B)**

2. Default Subnet Mask : 255.255.0.0 (/16)

3. Network (Subnet) Address : 142.251.0.0

4. **Broadcast Address** : 142.251.255.255

5. **Host Portion :** 216.46

6. **Number of Usable Hosts**

- Class B has 16 host bits:
- 2^16 − 2 =65,534 usable addresses
- **First Usable Host :** 142.251.0.1
- **Last Usable Host** : 142.251.255.254

Facebook.com - 102.132.101.35

1. IP Class : First octet: 102 (**Class A)**

2. Default Subnet Mask : 255.0.0.0 (/8)

3. Network (Subnet) Address : 102.0.0.0

4. **Broadcast Address** : 102.255.255.255

5. **Host Portion :** 132.101.35

6. **Number of Usable Hosts**

- Class A has 24 host bits:
- 2^24 − 2 =16,777,214 usable addresses

- **First Usable Host :** 102.0.0.1
- **Last Usable Host** : 102.255.255.254

X.com - 162.159.140.229

1. IP Class : First octet: 162 (**Class B)**

2. Default Subnet Mask : 255.255.0.0 (/16)

3. Network (Subnet) Address : 162.159.0.0

4. **Broadcast Address** : 162.159.255.255

5. **Host Portion :** 140.229

6. **Number of Usable Hosts**

- Class B has 16 host bits:
- 2^16 − 2 =65,534 usable addresses
- **First Usable Host :** 162.159.0.1
- **Last Usable Host** : 162.159.255.254

LinkedIn.com - 150.171.22.12

1. IP Class : First octet: 150 (**Class B)**

2. Default Subnet Mask : 255.255.0.0 (/16)

3. Network (Subnet) Address : 150.171.0.0

4. **Broadcast Address** : 150.171.255.255

5. **Host Portion :** 22.12

6. **Number of Usable Hosts**

- Class B has 16 host bits:
- 2^16 − 2 =65,534 usable addresses
- **First Usable Host :** 150.171.0.1
- **Last Usable Host** : 150.171.255.254

Wikipedia.org - 185.15.58.224

1. IP Class : First octet: 185 (**Class B)**

2. Default Subnet Mask : 255.255.0.0 (/16)

3. Network (Subnet) Address : 185.15.0.0

4. **Broadcast Address** : 185.15.255.255

5. **Host Portion :** 58.224

6. **Number of Usable Hosts**

- Class B has 16 host bits:
- 2^16 − 2 =65,534 usable addresses
- **First Usable Host :** 185.15.0.1
- **Last Usable Host** : 185.15.255.254

4.0 Conclusion

The ipv4 classification can be traced to their default subnets mask while the subnet mask holds the justification of network portion and host portion of an ip address. The class A ip address was found to be suitable for public or commercial applications while the class B ip address is suitable for medium scale ip address planning. The class C ip address is best for a SOHO network. Majority of the target IP addresses falls under class A and B private ip addresses as a result of the large scale enterprise size of the organisation using them.